Check for updates

Received: 4 May 2021 | Accepted: 29 March 2023

DOI: 10.1111/poms.14005

POMS

ORIGINAL ARTICLE

How incumbents beat disruption? Evidence from hotel responses to home sharing

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Handling Editor: Vijay Mookerjee

Funding information
National Natural Science Foundation of China, Grant/Award Numbers: 72002058, 72131005; Hong Kong Polytechnic University, Grant/Award Number: P0034898

Abstract

As the sharing economy continues to disrupt incumbent services, whether and how incumbents respond to the competition remains largely unknown. We investigate how incumbents can utilize management responses—a managerial intervention to guest reviews—to exploit performance improvement opportunities in guest reviews and sustain competitive advantage facing increased competition from home sharing. Our method integrates quasi-experiments, topic modeling, and deep learning techniques to not only estimate the impact of home sharing but also unveil the performance improvement mechanism. The findings reveal distinctive management response strategies across hotel price segments after home sharing's entry, which lead to divergent performance outcomes in guest satisfaction and sales. Regardless of their price segments, any hotel that responds more actively to guest reviews demonstrates improved guest satisfaction in service areas where home-sharing leads (e.g., check-in/out, cleanliness, sightseeing opportunity, and room conditions) and achieves higher sales. In contrast, hotels that respond less to guest reviews appear to lose guest satisfaction and sales to not only home sharing but also peer hotel cohorts that respond more. Our study contributes to the literature on the intersection of service operations and technology and provides timely implications that can inform incumbents' response strategies to disruptions in the ever-changing business world.

KEYWORDS

deep learning, difference-in-differences, incumbent firms, management responses, sharing economy

1 | INTRODUCTION

Sharing economy platforms such as Airbnb and Uber have empowered individuals to offer services that used to be provided solely by firms (hotels and taxis). In particular, the rise of home-sharing services such as Airbnb has increasingly enticed guests out of hotel rooms to local residential homes with elevated guest experience (Farronato & Fradkin, 2022; Li & Srinivasan, 2019; Zervas et al., 2017, 2021). The intensified competition is evident in the fact that Airbnb has outperformed all major hotel chains (Marriott, Hilton and IHG, to name a few) by room supply and valuation as of 2022 (Ballard, 2022). Such success has made home sharing a travel industry disruptor, pushing incumbent hotels to rethink the way they offer services. Amid the disruption of sharing economy is the unaddressed curiosity about whether and how incumbent firms battle back. The existing literature is silent on a few questions of particular interest to academics and practitioners: Whether and how do incumbents respond to disruptors? Is their response strategy effective? What is the mechanism through which incumbents can beat disruptors? These questions motivate our study and are timely relevant to incumbent firms as well as sharing economy disruptors.

In this paper, we study incumbent hotels and their responses to home sharing. While the literature documents price responses as a primary incumbent response (e.g., McCann & Vroom, 2010; Seamans, 2013; Simon, 2005), we focus on a non-price strategy: quality improvement utilizing management responses to guest reviews. In the form of an open-ended piece of text, a management response is written by hotel managers and publicly displayed underneath guest reviews (reviews, hereafter) that it aims to address. Literature has widely documented management responses as an effective strategy to collect feedback from guests and

Accepted by Vijay Mookerjee, after two revisions.

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improve service quality (Gu & Ye, 2014; Khurana et al., 2019; Wang & Chaudhry, 2018). Yet, whether such a strategy remains effective in addressing the disruption from home sharing remains unknown. We empirically examine the change in management responses across incumbent hotels after home sharing's entry, how divergent response strategies differentiated hotels in performance, and what quality improvement opportunities were found from responding to reviews.

We collected data on hotels and home sharing in a primary tourist market in China over a period from 2015 to 2017. During this period, the tourist market experienced exponential growth in home sharing, which allows us to observe the evolution of competition between hotels and home sharing. We obtained data on purchase-verified reviews, management responses to these reviews, as well as property location (geographic coordinates) and characteristics of hotels. For each hotel, we also identify home sharing's staggered entry to its vicinity and calculate home sharing supply accordingly. We combine the data from multiple sources and construct a panel at the hotel by year-month level. Our final sample includes 6678 hotels and 3584 home-sharing properties.

The variations in the timing of home sharing's entry into the vicinity of hotels represent an empirical opportunity to observe hotels' responses to disruption. Our estimation hinges on a difference-in-differences (DID, hereafter) design, in which hotels experiencing home sharing's entry is the affected group, whereas those not yet experiencing the entry are the control group. Because our data are rich in unstructured textual reviews on both hotels and their home-sharing counterparts, we also utilize topic modeling and deep learning techniques to observe the quality improvement evidence in hotels that beat the disruptors. Specifically, our algorithms integrate topic modeling via latent Dirichlet allocation (LDA, hereafter), deep learning for sentiment analyses via Word2Vec, and convolutional neural networks (CNN, hereafter). We begin by training an integrated model to obtain the detailed topics of hotel reviews with corresponding sentiments on each topic. We then replicate the analyses on home-sharing reviews and compare reviews between hotels and home sharing to identify quality improvement evidence by service area. We also conduct a battery of tests to rule out alternative explanations and check the robustness of our results. A summary of these tests is reported in Table W1 of the Web Appendix.

Our analyses glean important insights. We first find a sharp divergence between higher-priced hotels and lower-priced hotels in their management response activities, although the average hotel management responses remain stable. Specifically, management responses to reviews surge by 2.7% at higher-priced hotels, while plummeting by 3.2% at lower-priced hotels. Such distinction indicates the heterogeneity of incumbent responses across hotel price segments when facing home-sharing disruption.

A natural question follows: What is the impact of asymmetric response activities on hotel performance? We investigate the change in guest satisfaction and sales across hotel price segments after home sharing's entry. Interestingly, we find that higher-priced hotels enjoy a significant increase in both satisfaction and sales. In contrast, lower-priced hotels experience a drop in satisfaction and sales. Although lower-priced hotels overall respond less to reviews, those striving to respond more gain higher performance. That is, hotels in any price segment could improve their performance if proactively responding to reviews. It appears to be a competitive advantage for hotels to respond to reviews, which is not only a response strategy to disruptors but also a differentiation strategy to other hotel cohorts.

What is the mechanism behind the improved performance of hotels that actively respond to reviews after home sharing's entry? We exploit the richness in our data to unveil the content features (topics and sentiments) of review texts for both hotels and home sharing. Because each review is essentially "a bag of topics," we use topic modeling techniques to extract theme-specific topics from the sentences. We then utilize deep learning's Word2Vec and CNN algorithms to evaluate the sentiment associated with each topic. By comparing the topics yielded from hotel reviews and home-sharing reviews, we find that four areas (cleanliness, the check-in/out process, room conditions, and sightseeing opportunities) speak to the quality gaps between hotels and home-sharing. Next, we formally test whether responding to these review topics boosts hotel performance after home sharing's entry. We model the impact of entry on the sentiments of all review topics and find that hotels actively respond to reviews and indeed obtain higher sentiments on these four topics. It is evident that management responses to reviews on cleanliness, sightseeing opportunities, room conditions, and check-in/out process help hotels improve guest satisfaction against the disruption of home sharing.

Our paper contributes to the literature on sharing economy, entry and incumbents, and service operations. While the literature on sharing economy primarily documents its impact on incumbent firms (Farronato & Fradkin, 2022; Li & Srinivasan, 2019; Pan & Qiu, 2022; Zervas et al., 2017, 2021; Zhou & Wan, 2022), we add a less studied but important perspective on how incumbents could proactively respond to the disruption. Complementing the literature on entry and incumbents (McCann & Vroom, 2010; Prince & Simon, 2015; Pei et al., 2021; Seamans, 2013; Simon, 2005), we advocate a non-price strategy (management responses) with quality improvement effectiveness for incumbents to address the competition. As management responses are a stark example of service operations, our study adds to the literature on how to use online managerial intervention to collect guest feedback and identify areas for quality improvement in hotel operations (Gu & Ye, 2014; Wang & Chaudhry, 2018). Our hybrid method integrating quasi-experiment, topic modeling, and deep learning techniques lies at the intersection of operations management and service management. For incumbents who face disruption, we advocate a guest-centric approach to exploit reviews and actively respond to the reviews for performance improvement opportunities and sustainable

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competitive advantage. Through management responses, different types of incumbents can extrapolate actions specific to quality improvement, as service quality is arguably the most important factor for gaining a competitive advantage (Oh & Kim, 2017). Specific to the hotel industry practitioners, these actions include adjusting service quality to bridge the gap with disruptors in areas such as the check-in/out process, cleanliness, sightseeing opportunity, and room conditions.

The rest of the paper is organized as follows. First, we review relevant literature and discuss our contribution to existing research. We then introduce the data and measures. We present analyses and findings of quasi-experiment, topic modeling, and deep learning techniques, along with additional analyses to rule out alternative explanations and check the robustness of results. We conclude the paper by discussing its implications.

2 | RELATED LITERATURE

Our study is related to the seminal work in home sharing, entry and incumbents, and management responses. Our major contributions lie primarily in three important areas: (1) understanding management responses are an essential service operation strategy when disruptors challenge the incumbent businesses, (2) identifying what aspects of services shall be responded to for performance improvement opportunities against disruption, and (3) how divergent response strategies would make a difference in performance between different incumbents. We review related literature and discuss the contributions in this section.

2.1 | Home sharing

This paper is first related to the literature on home sharing. Research on the rise of home sharing primarily focuses on how home sharing disrupts incumbent counterparts that offer similar services. On home sharing's impact on incumbent hotels, literature documents how home sharing cut into hotels' profitability (Farronato & Fradkin, 2022; Li & Srinivasan, 2019; Zervas et al., 2017). On home sharing's impact on housing markets, research shows how home sharing competes with local housing and rental markets for home supply (Barron et al., 2021; Chen et al., 2022; Horn & Merante, 2017).

Despite much being known about home sharing's impact on incumbents, less is known about incumbents' responses to home sharing. Wallsten (2015) investigate how the incumbent taxi industry reduces customer complaints in response to the growing popularity of Uber. Chang and Sokol (2020) examine the price and non-price response strategies of incumbent hotels to Airbnb and found that lower-quality hotels compete on price and higher-quality hotels reposition themselves in the higher end of the lodging market. We extend both studies in unveiling how non-price, quality improvement strategy can effectively drive incumbent performance in facing disruption. We contribute a unique approach to management responses to address the goal of quality improvement against competition and enrich the findings of previous studies.

2.2 | Entry and incumbents

Our study contributes to the literature on entry and incumbents. Traditional industrial organization studies (e.g., see Tirole, 1988) discuss two main strategies to respond to entrants. The first is collusive behavior (either coordinated or tacit), which is often employed in the early stages of the category life cycle when there are few competitors (Stigler, 2021). The main idea is that, when the number of firms in a category is small, it is easier for firms to coordinate their actions (Chevalier-Roignant et al., 2019; Sutton, 1974). The second is the general marketing spending, which performs the expected role of increasing the number of customers. Such a strategy is to increase the difficulty of generating awareness and trial for a new product, especially when growth is accelerated by new firms that enter the market (Xiang et al., 2021).

Our research context and the response strategies to entry are different from classic industrial organization research. First, incumbent hotels are a more mature industry facing disruptions, while home sharing as an emerging disruptive business model is also different from traditional entrants. Second, we advocate responding to guest reviews for performance improvement opportunities as an incumbent response strategy and demonstrate the effectiveness of such a strategy, which is quite different from the price tactics in traditional literature contexts. We show how incumbents can utilize management responses to exploit performance improvement opportunities in guest feedback and sustain competitive advantage. By showing evidence of how incumbents adjust their quality provision in response to disruptors, we suggest an effective strategy (management response) to the literature and demonstrate its value to incumbents facing a competitive move such as entry by rivals.

2.3 | Management responses

The literature on management responses shows that firms have strong incentives to engage consumers online (Xie et al., 2014), learn from the "wisdom of crowds" (Park & Allen, 2013), and determine what they can do to meet customer needs (Kumar et al., 2018). As an important online reputation management strategy, management responses can improve online reputation proxied by the valence of subsequent reviews (Chevalier et al., 2018; Gu & Ye, 2014; Wang & Chaudhry, 2018), stimulate customer engagement by soliciting customer opinion manifested in the larger volume of subsequent reviews (Chen et al., 2019; Proserpio & Zervas, 2017), and eventually drive firm performance (Kumar et al., 2018). Studies investigating management responses have

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explained the role of such responses from various theoretical perspectives, including service recovery (Dens et al., 2015; Kim et al., 2015; Lee & Song, 2010; Xie et al., 2014), the accessibility-diagnosticity model (Lee et al., 2017), and consumer trust (Crijns et al., 2017; Marx & Nimmermann, 2017; Mauria & Minazzi, 2013; Sparks et al., 2016; Wei et al., 2013).

We add to the literature by advocating another value of management responses: identifying opportunities to beat disruptors. Unveiling the mechanism behind the impact of management responses, we provide a clear picture of what quality improvement opportunities hotel managers identify from reviews after home sharing's entry and how they resort to the opportunities to improve performance and compete with the rivals. Our quasi-experiment, combined with the topic modeling and deep learning approach, identifies the change in the number of management responses as a measure of manager reactions to disruption and drills down to specific service areas that drive performance improvement from the review texts. This represents a contribution as our findings unveil precise areas that the incumbents focus on when facing increased disruption. The result sheds light on the new role of management responses: performance improvement amid intensified competition.

2.4 | Topic modeling and deep learning on review texts

Research using unstructured data such as reviews has flourished over the past few years (Archak et al., 2011; Decker & Trusov, 2010; Lee & Bradlow, 2011; Tirunillai & Tellis, 2014). Yet, mining unstructured textual data using either hand-coded features or intensive feature engineering can be ad hoc and error-prone (Liu et al., 2019). The large volume of reviews in the big data era makes human coding time-consuming. We join a few pioneer studies (e.g., Liu et al., 2019; Timoshenko & Hauser, 2019; Zhang & Luo, 2022) to adopt topic modeling and deep learning techniques to extract the contents from textual reviews and achieve scalability. Specifically, we rely on topic modeling (Blei et al., 2003) and deep learning (LeCun et al., 2015) that can use raw data to automatically discover feature representations and process large-scale, unstructured information at the highly granular level. To extract the topics and sentiments of review texts simultaneously, our approach integrates topic modeling and deep learning using sentences as the basic unit of deep learning and achieves good performance in the validation process. As deep learning techniques continue to prevail, we make a meaningful step in advocating algorithmic analyses for business research.

3 | DATA

We collect data from Beijing, a primary tourist market in China. This accommodation market witnessed an influx of home sharing in local hotel markets between 2015 and 2017. Our data on hotels are collected from Ctrip (ctrip.com), the largest online travel agent in China.[1] For each hotel in Beijing, we collected its purchase-verified reviews (ratings and texts) as well as the hotel managers' management responses underneath reviews. Our data on home sharing are collected from Xiaozhu (xiaozhu.com). Known as "Airbnb in China," Xiaozhu is the largest home-sharing platform in China. As of December 2017, Xiaozhu has more than 8 million home-sharing properties in over 400 domestic cities.[2] We identify home sharing's entry by observing its presence within a 500-m radius of a hotel.[3] We label the entry of each home-sharing property using the date of its first purchase-verified guest review published on the platform. Following Zervas et al. (2017), we measure home sharing's entry utilizing two measures: the entry dummy (*Entry*) and the logarithm of the cumulative supply (*logSupply*) of home-sharing properties from the entry month up to the current month. Table 1 presents the variable definitions and summary statistics.

For each hotel, we calculate the average rating of hotel reviews (a proxy for guest satisfaction or *Satisfaction*) and count the number of these purchase-verified reviews (a proxy for sales or *logSales*[4]) in a month. We measure hotel management responses using *MRRatio*, which is the ratio of the number of management responses to the number of hotel reviews in a month. We also collect data on hotel characteristics, including the cumulative average rating (*Rating*) and the cumulative reviews (*logReviews*) of hotels, the number of other hotels in a hotel's vicinity (*logHotels*), the indicator of whether a hotel's nightly rate is higher than the median nightly rate of all hotels in the sample (*HighPrice*), and perceived hotel price based on guest sentiments on hotel price identified in their hotel reviews (*PerceivedPrice*).[5] We take logarithms of several variables that exhibit skewed distributions. Therefore, some of our estimations will inherit the semi-elasticity interpretation.

We combine data from different sources into a panel at the hotel by year-month level. Our sample includes 6678 hotels and 3584 home-sharing properties over a period from March 2015 to December 2017 (a total of 34 months). Figure 1 presents the price distribution of both home sharing and hotels (per bed) in our sample. It appears that the overlap between the two occurs on every single price level. A majority of the price overlap concentrates on the price range of RMB100—RMB500, in which home sharing has a slightly larger share on the higher bound.

Table 2 exhibits the difference between hotel price segments (higher-price hotels and lower-price hotels) by their management responses and performance outcomes, such as guest satisfaction and sales. The result provides preliminary evidence of the heterogeneity among hotel price segments when facing the entry of home sharing: It seems higher-priced hotels experience an increase in management responses after home sharing's entry, and as a result, so do guest satisfaction and sales, while lower-priced hotels show otherwise.

TABLE 1 Variable definitions and summary statistics

	Definition	Mean	Median	SD	Min	Max
Hotel performance						
Satisfaction	Average rating of purchase-verified hotel reviews	4.214	4.380	0.713	1	5
logSales	Logarithm of the number of purchase-verified hotel reviews[a]	2.073	2.079	1.400	0	6.688
Home sharing's entry						
logSupply	Logarithm of the cumulative number of home-sharing properties that have entered a hotel's vicinity	0.990	0.693	1.105	0	4.754
Entry	Dummy variable indicating whether home sharing has entered a hotel's vicinity	0.641	1	0.480	0	1
Management responses of hotels						
MRRatio	Ratio of the number of management responses to the number of hotel reviews	0.382	0	0.464	0	1
Hotel characteristics						
Rating	Cumulative average rating of hotel reviews as of last month	4.190	4.275	0.482	1	5
logReviews	Logarithm of the number of cumulative reviews as of last month	4.493	4.700	1.823	0	9.265
logHotels	Logarithm of the number of other hotels in a hotel's vicinity	2.239	2.398	1.015	0	4.574
HighPrice	Dummy variable indicating whether a hotel's nightly rate is higher than the median nightly rate of all hotels in our sample	0.495	0	0.500	0	1
PerceivedPrice	Perceived price based on guest sentiments identified in their price-related hotel reviews	0.367	0.25	0.400	0	1

[a]Not all guests who have stayed at a hotel would leave a review, and only one review is allowed for each verified stay. Thus, our definition of sales implies a conservative estimate of the impact of management responses.

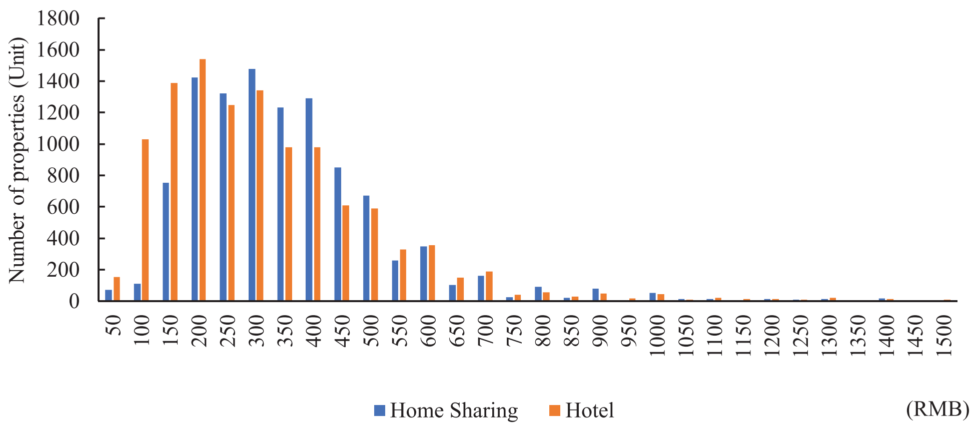

FIGURE 1 Comparing price distribution of hotels and home sharing (normalized price per bed). [Color figure can be viewed at wileyonlinelibrary.com]

4 | ANALYSES AND FINDINGS

As a quick roadmap, we begin with identifying home sharing's impact on hotel management responses and performance, with a focus on the heterogeneity across hotel price segments. We attribute such performance divergence to the different hotel management responses. To discover the underlying mechanism behind the use of management responses, we analyze review texts using topic modeling and deep learning to identify service gaps between hotels and home sharing. We nail down the mechanism that management responses to these reviews help bridge the service gaps between hotels and home sharing. The last two sections present additional analyses to rule out alternative explanations and check the robustness of the findings.

4.1 | Impacts of home sharing on hotels' management responses and performance

The vast variations in the temporal rate of home sharing's geographical expansion across hotel vicinities provide a quasi-experimental opportunity. By the end of our study period, there were 4432 hotels experiencing home sharing's

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TABLE 2 Differences in management responses and performance between hotel price segments

	(1)	(2)	(3)
	MRRatio	***Guest Satisfaction***	***logSales***
Panel 1. Higher-priced Hotels			
Before entry	0.479	4.320	2.403
After entry	0.482	4.360	2.684
Panel 2. Lower-priced Hotels			
Before entry	0.299	4.125	1.433
After entry	0.275	4.051	1.342

entry (the treatment group) and 2246 hotels not experiencing home sharing's entry (the control group). Our specification is a regression-adjusted DID model (Angrist & Pischke, 2008).[6] For each hotel i in year-month t, its management responses are a function of home sharing's entry and other covariates. The equation is as follows:

$$Y_{it} = \beta_0 \cdot Treated_i + \beta_1 \cdot Treated_i \times logSupply_{it} + \mu_i + \nu_t + w_{it} + \delta' \cdot Z_{it} + \varepsilon_{it}, \quad (1)$$

where Y_{it} is the dependent variable. Depending on the specific regression, we examine the management response activities and hotel performance outcomes as a result of home sharing's entry. $Treated_i$ indicates whether a hotel has experienced home sharing's entry by the end of the study period. Following the literature (Zervas et al., 2017), we use the logarithm of the number of home-sharing properties that have entered the 500-m radius of the hotel i in year-month t, $logSupply_{it}$, as the treatment.[7] We consider a 500-m radius[8] as a valid hotel vicinity to observe the market competition between hotels and home sharing. Because $logSupply_{it,}$ is both hotel and year-month specific, $Treated_i \times logSupply_{it}$ is equivalent to $logSupply_{it}$.[9] Note that $Treated_i$ will be absorbed by the hotel fixed effects.

Z_{it} represents a set of control variables, including *logHotels, Rating, logReviews*, and *PerceivedPrice*. We also control $logHotels_{it}$ because, arguably, the agglomeration and competition from peer hotels are associated with a hotel's response to disruption and its subsequent performance. Controlling the cumulative historical hotel performance $Rating_{it\text{-}1}$ and $logReviews_{it\text{-}1}$ is to account for any inherent quality difference of the hotels. Note that we lagged these two variables to avoid including the current values of the dependent variables. $PerceivedPrice_{it}$ is controlled to represent the time-variant perceived hotel price based on guest sentiments. We include hotel fixed effects, μ_i, year-month fixed effects, ν_t, and w_{it}, the district-specific trends, calculated by interacting district dummies and month trends. The residual term, ε_{it}, captures idiosyncratic shocks.

TABLE 3 Impact of home sharing on hotel management responses.

Dependent Variable (D.V.): *MRRatio*	**(1)**	**(2)**	**(3)**
logSupply	0.001	−0.032***	−0.305***
	(0.002)	(0.003)	(0.012)
logSupply × *HighPrice*		0.059***	
		(0.003)	
logSupply × *logPrice*			0.055***
			(0.002)
Control Variables			
logHotels	−0.038***	−0.041***	−0.039***
	(0.007)	(0.007)	(0.006)
Rating	−0.004***	−0.002**	−0.002**
	(0.001)	(0.001)	(0.001)
logReviews	0.007***	0.004***	0.003***
	(0.001)	(0.001)	(0.001)
PerceivedPrice	0.010***	0.009***	0.009***
	(0.002)	(0.002)	(0.002)
Hotel Fixed Effects (FE)	YES	YES	YES
Month Fixed Effects (FE)	YES	YES	YES
District-specific trends	YES	YES	YES
Observations	118,111	118,111	118,111
R-squared	0.706	0.708	0.708

Note: Clustered robust standard errors in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

4.1.1 | Impact of home sharing on hotel management responses

We first test whether and how hotel management responses change when facing home sharing's entry. Table 3 reports the results based on our main specification. The dependent variable is the ratio of management responses to guest reviews, *MRRatio*. First, we find no significant association between home sharing's entry and management responses of all hotels as shown in column 1. On average, hotels appear not responsive to home sharing's entry. We further incorporate the heterogeneity of hotels (by price segment) by interacting *logSupply* with *HighPrice*, a dummy variable indicating a higher-priced hotel based on whether its price is above the median of all hotels. The result exhibits asymmetric response strategies between hotel price segments. As column 2 shows, we find a significant surge in management responses of higher-priced hotels after home sharing's entry, whereas the management responses plummet in lower-priced hotels. Numerically, a 1% increase in home-sharing properties leads to a 2.7% increase in management responses of higher-priced hotels but a 3.2% drop in lower-priced hotels. Column 3 adopts a continuous *logPrice* variable as the moderator, and we also find that the management responses increase more for hotels with higher prices.

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TABLE 4 Impact of home sharing on hotel guest satisfaction and sales

	(1)	(2)	(3)	(4)
D.V.s:	***Satisfaction***		***logSales***	
logSupply	−0.003	−0.019***	−0.011**	−0.037***
	(0.004)	(0.006)	(0.005)	(0.006)
logSupply × *HighPrice*		0.027***		0.046***
		(0.006)		(0.006)
Control Variables				
logHotels	0.014	0.012	−0.040***	−0.043***
	(0.013)	(0.013)	(0.014)	(0.014)
Rating	−0.020***	−0.020***	0.079***	0.081***
	(0.003)	(0.003)	(0.002)	(0.002)
logReviews	0.011***	0.009***	0.039***	0.036***
	(0.003)	(0.003)	(0.003)	(0.003)
PerceivedPrice	0.028***	0.028***	0.391***	0.391***
	(0.005)	(0.005)	(0.005)	(0.005)
Hotel FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
District-specific trends	YES	YES	YES	YES
Observations	118,111	118,111	118,111	118,111
R-squared	0.350	0.351	0.829	0.829

Note: Clustered robust standard errors in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

4.1.2 | Impact of home sharing on hotel guest satisfaction and sales

After observing the divergent management response strategies across hotels, a natural question is whether such divergence leads to different performance outcomes. We next examine the impact of home sharing on guest satisfaction and sales of hotels after home sharing's entry. We present the estimation results in Table 4. As columns 1 and 3 show, the coefficient of *logSupply* indicates that home sharing does negatively impact guest satisfaction and sales to all hotels, even though the effect on guest satisfaction is not statistically significant.

In addition to the all-hotel analysis, we also conduct analyses by hotel price segments. The coefficients of the *logSupply* × *HighPrice* interaction in columns 2 and 4 reveal surprising impacts of home sharing on guest satisfaction and sales, respectively, between two hotel price segments. Similar to what we find in Table 3 for management responses, column 2 shows that guest satisfaction with higher-priced hotels increases significantly, while that of lower-priced hotels decreases after home sharing's entry. Quantitively, as home-sharing properties increase by 1% in a lower-priced hotel's vicinity, its guest satisfaction decreases significantly by about 0.019 out of 5-star ratings. Oppositely, a 1% increase in home-sharing properties around a higher-priced hotel drives its guest satisfaction by 0.008 (= 0.027 – 0.019). The results of hotel sales exhibit a similar pattern in column 4. A 1% increase in home-sharing properties in a lower-priced hotel's vicinity is associated with a 0.037% decrease in sales. This finding is completely in line with Zervas et al.'s (2017) estimates on the revenue cut of hotels after home sharing's entry. Yet, for higher-priced hotels, we find the sales, intriguingly, increase by 0.009% (=0.046 – 0.037) against home sharing disruption.

Taken together, these divergent changes in guest satisfaction and sales between hotel price segments echo their different strategies in reacting to home-sharing disruption: Higher-priced hotels tend to serve guests better and make more sales after home sharing's entry, while lower-priced hotels sink in guest satisfaction as well as sales. We may attribute this result to the divergence in hotels' management response strategies for dealing with competition from home sharing's entry, which we test empirically in the next section.

4.1.3 | The driver of performance divergence: Management responses

To examine the mechanism behind the difference in hotel performance, we test whether such a difference is driven by management responses (rather than other hotel characteristics such as price segment). We examine whether a higher level of management response activities indeed leads to higher guest satisfaction and sales after home sharing's entry. To this end, we develop two variables to measure the changes in hotel management responses: *MRRatioDiff* and

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TABLE 5 Impact of home sharing on hotel guest satisfaction and sales: moderated by response quantity

	(1)	(2)	(3)	(4)
	Satisfaction		*logSales*	
D.V.s:	**Higher-priced**	**Lower-priced**	**Higher-priced**	**Lower-priced**
logSupply	−0.003	−0.012	−0.011	−0.043***
	(0.005)	(0.008)	(0.007)	(0.007)
logSupply × *MRRatioDiff*	0.006	0.019	0.062***	0.047***
	(0.006)	(0.011)	(0.010)	(0.013)
Control Variables				
logHotels	−0.014	0.042*	−0.004	−0.077***
	(0.014)	(0.021)	(0.020)	(0.021)
Rating	−0.018***	−0.020***	0.109***	0.064***
	(0.003)	(0.004)	(0.004)	(0.003)
logReviews	0.007**	0.010**	0.052***	0.020***
	(0.003)	(0.004)	(0.004)	(0.004)
PerceivedPrice	0.016***	0.038***	0.367***	0.402***
	(0.006)	(0.007)	(0.008)	(0.007)
Hotel FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
District-specific trends	YES	YES	YES	YES
Observations	58,448	59,663	58,448	59,661
R-squared	0.362	0.314	0.843	0.753

Note: Clustered robust standard errors in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

MatchedMRRatioDiff.[10] The former measures the average difference of management response ratios before and after home sharing's entry for each hotel (a response *quantity* measure), while the latter measures the average difference in the ratios of management responses that are more relevant to guest reviews before and after home sharing's entry for each hotel (a response *quality* measure). We interact *logSupply* with these two measures and replicate the analyses.[11] If it is the management response strategy, rather than hotel characteristics such as the price segment, that drives guest satisfaction and sales, we should expect a positive and significant coefficient for the interaction term. To incorporate the heterogeneity of hotels, we estimate the interaction term for both higher-priced hotels and lower-priced hotels, consistent with the previous analyses.

Table 5 presents the results for *MRRatioDiff*. Columns 1 and 2 report the moderating effect of *MRRatioDiff* on guest satisfaction. We find that home sharing negatively affects hotel guest satisfaction, and the hotel has an increase (yet not significant) in satisfaction if it has a larger difference in management response ratio before and after home sharing's entry for both the higher-priced and lower-priced hotels.[12] Columns 3 and 4 show that home sharing has an especially significant negative impact on the sales of lower-priced hotels than that of higher-priced hotels. Moreover, the moderating effects of *MRRatioDiff* are positive and significant, implying hotels would have attenuated the sales drop after home sharing's entry if they actively responded to guest reviews.

Table 6 exhibits the results for *MatchedMRRatioDiff*. Much similar to Table 5, columns 1 and 2 show the moderating effect of the difference in the ratios of management responses with close relevance to guest reviews before and after home sharing on guest satisfaction. Columns 3 and 4 show that the moderating effects of *MatchedMRRatioDiff* are also significantly positive, with even a slightly higher coefficient magnitude than that for the moderating effect of *MRRatioDiff* (the response quantity measure in Table 5). The reason may be that

Management responses with higher relevance to guest reviews reflect more managerial efforts in addressing guest reviews by hotel managers and, as a result, more opportunities for performance improvement. Such relevance is valuable, especially when extensive management responses read mechanically standardized and repetitive these days (Zhang et al., 2020). Overall, our results suggest that, when facing home sharing's entry, performance improvement opportunities are available if hotel managers actively respond to reviews. Their responses are effective in elevating sales against the increased competition from home sharing. Hotels that are more active in responding to guest reviews are more easily to seize the opportunities. Next, we examine what exactly more responsive hotels have learned from guest

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TABLE 6 Impact of home sharing on hotel guest satisfaction and sales: Moderated by responses quality

	(1)	(2)	(3)	(4)
	Satisfaction		*logSales*	
D.V.s:	Higher-priced	Lower-priced	Higher-priced	Lower-priced
logSupply	−0.001	−0.011	−0.004	−0.041***
	(0.005)	(0.008)	(0.007)	(0.007)
logSupply × *MatchedMRRatioDiff*	0.005	0.026	0.229***	0.216***
	(0.022)	(0.053)	(0.038)	(0.058)
Control Variables				
logHotels	−0.014	0.042*	−0.003	−0.077***
	(0.014)	(0.021)	(0.020)	(0.021)
Rating	−0.018***	−0.021***	0.109***	0.064***
	(0.003)	(0.004)	(0.004)	(0.003)
logReviews	0.007**	0.011**	0.052***	0.020***
	(0.003)	(0.004)	(0.004)	(0.004)
PerceivedPrice	0.016***	0.038***	0.367***	0.402***
	(0.006)	(0.007)	(0.008)	(0.007)
Hotel FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
District-specific trends	YES	YES	YES	YES
Observations	58,448	59,663	58,448	59,661
R-squared	0.362	0.314	0.843	0.753

Note: Clustered robust standard errors in parentheses.
***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

reviews using topic modeling and deep learning on review texts.

4.2 | Roles of hotel management responses against home sharing disruption

We have shown that management responses by managers can elevate hotel performance in terms of guest satisfaction and sales when facing home-sharing disruption. In this section, we adopt topic modeling and deep learning to understand what performance improvement opportunities are identified by hotel managers responding to reviews against home-sharing disruption.

4.2.1 | Identifying service gaps between hotels and home sharing

We process both hotel reviews and home-sharing reviews, each structured differently by two platforms (Ctrip and Xiaozhu). While hotel reviews are structured with a piece of review text and the associated numeric overall rating and sub-ratings, home-sharing reviews are solely textual without any numeric ratings or sub-ratings to indicate review sentiments. Figure W1 in the Web Appendix compares the guest review exhibition on two platforms (Ctrip for hotels and Xiaozhu for home sharing). To process the two sources of guest review data with consistent approaches, we first train the topic modeling and deep learning models to identify commonly reviewed topics in hotels and their associated guest sentiments. We then use the trained model to further classify commonly reviewed topics of home-sharing reviews and their associated sentiments. Comparing topics and sentiments of both hotels and home sharing, we reveal insights about service areas that strike the competition areas between hotel incumbents and home sharing. Section W2 in the Web Appendix reports technique details of guest review analyses.

Table 7 shows seven areas commonly reviewed by guests and their associated sentiments between hotels and home sharing. For easy reference, we denote the seven topics as T1–T7. The findings suggest the check-in/out process (T1), cleanliness (T3), sightseeing opportunity (T4), and room conditions (T5) are the top four areas where home-sharing reviews exhibit higher sentiments (guest satisfaction) than hotel reviews. These four topics point to quality adjustment and performance improvement opportunities for hotel managers. Next, we investigate whether these areas have been effectively addressed by hotel managers using management responses and whether responding to reviews in these areas improves hotel performance. Such insights will inform mechanisms behind the performance divergence among hotels adopting different management response strategies amid home-sharing disruption.

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TABLE 7 Comparing review sentiments on service areas between home sharing and hotels

	Review sentiments		
	(1)	(2)	(3)
Review topics	Hotels (Mean)	Home sharing (Mean)	Sentiments difference
Topic 1 (T1): Check-in/out process[a]	0.408	0.71	0.302
Topic 2 (T2): Facility and amenity	0.692	0.922	0.23
Topic 3 (T3): Cleanliness[a]	0.331	0.651	0.32
Topic 4 (T4): Sightseeing opportunity[a]	0.56	0.807	0.247
Topic 5 (T5): Room conditions[a]	0.428	0.716	0.288
Topic 6 (T6): Location	0.771	0.894	0.123
Topic 7 (T7): Guest service	0.876	0.953	0.077

[a]indicates service areas on which reviews from home sharing exhibit higher sentiments than reviews from hotels.

4.2.2 | Improving service gaps using hotel management responses

To identify the roles of management responses in hotel performance improvement after home sharing's entry, we incorporate the seven topics and their deep learning-detected sentiments into Equation (1), where we replace Y_{it} using the topic sentiments. Our conjecture is, if management responses are effective in driving hotel performance against home-sharing disruption, we should see a significant quality improvement in service areas (reviewed topics) that experience management responses. The results are presented in Table 8. As we can see, after home sharing's entry, hotels responding less to reviews do not see a significant change (with non-significant coefficients of log*Supply*) in review sentiments on almost all topics except for guest service in general. In sharp contrast, hotels that have adopted a proactive response strategy (with a larger difference in management response ratio before and after home sharing's entry), the sentiments on the check-in/out process (T1), cleanliness (T3), sightseeing opportunity (T4), and room conditions (T5) have increased significantly. Note that the sentiments are between 0 and 1. The magnitude of the coefficients is, therefore, small. These four topics are exactly the service areas representing quality gaps between hotels and home sharing, as shown earlier in Table 7. The improvement in other dimensions is mostly positive, though not statistically significant. The results demonstrate that hotels actively responding to reviews indeed experience improved sentiments (guest satisfaction) on service areas that fall behind home sharing.

One may question how effective management responses are in reducing negative reviews in service areas that fall behind home sharing (i.e., T1, T3, T4, and T5). Evidence addressing this question will provide further insights into the role of hotel management responses in alleviating home-sharing disruption. The conjecture is that, after hotel managers respond to reviews on these seven service areas, there is a significant decrease in their subsequent negative reviews; for service areas that do not receive any management responses, there is no statistically significant decrease in their subsequent negative reviews. To test the conjecture, we generate the logarithm of the number of negative reviews in each topic as the dependent variable (i.e., log*NegativeReviews on Topic Tn*, where $Tn \in \{T1, \dots, T7\}$) and calculate the ratio of management responses to reviews on each topic in the last month (i.e., *Tn MR Ratio* (t-1)) as the independent variable. Table 9 exhibits the regression results. We can find significant decreases in negative reviews across seven topics (service areas) after receiving management responses. The findings indicate that future concerns (negative reviews) about these seven areas are significantly alleviated after hotel managers actively respond to previous reviews on these areas.

Following the same specification, we develop a binary measure indicating the absence of management responses (i.e., *Tn No MR* (t-1)), with values of 1 being no management responses to reviews on a given topic in the last month and 0 otherwise. We replace *Tn MR Ratio* ($t-1$) in Table 9 using the binary measure *Tn No MR* ($t-1$) and re-run the regression. The results are shown in Table 10. We find service areas without management responses experience insignificant change. Taking the results of Tables 9 and 10 together, it is evident that management responses can effectively improve quality in service areas where hotels fall behind home sharing; without actively responding to reviews, it is challenging for hotels to improve service quality and bridge the quality gap with home sharing.

4.3 | Additional analyses

In this section, we provide additional analyses to rule out alternative explanations and check the robustness of the findings.

4.3.1 | Potential change in the reviewer base

Some may worry that complaining guests at hotels may have switched to home sharing, and the loss of negative reviews (by these complaining guests) may have driven hotel guest satisfaction and confounded the impact of management responses. We argue that whether complaining guests at hotels reduce after home sharing's entry is an empirical question. The alternative explanation (a decrease in complaining reviewers), if true, would predict an increase in the average review rating of hotels after home sharing's entry due to the loss of reviewers who write negative reviews. Such an increase should be larger in lower-priced hotels because they were affected more by home-sharing disruption (Zervas et al., 2017). We have tested whether the average rating of hotels, especially the lower-priced ones, indeed changes after home sharing's entry. The results by hotel price segments have been reported in Table 4, which suggests that lower-priced hotels experience a significant decrease in guest satisfaction after

TABLE 8 Impact of home sharing and hotel management responses on review topic sentiments

D.V.:	(1) T1 Check-in/out	(2) T2 Facility and amenity	(3) T3 Cleanliness	(4) T4 Sightseeing experience	(5) T5 Room conditions	(6) T6 Location	(7) T7 Guest service
logSupply	−0.002	0.001	−0.001	−0.001	−0.000	0.000	0.004**
	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.002)	(0.002)
logSupply MRRatioDiff*	0.012***	−0.001	0.014***	0.009**	0.007*	0.003	−0.001
	(0.004)	(0.004)	(0.005)	(0.005)	(0.004)	(0.002)	(0.002)
Control Variables							
logHotels	0.018**	0.015*	−0.006	0.000	−0.001	0.008	−0.005
	(0.008)	(0.008)	(0.011)	(0.010)	(0.008)	(0.005)	(0.004)
Rating	−0.004**	−0.001	−0.002	0.000	0.002	0.001	−0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
logReviews	0.002	−0.004**	−0.003	−0.000	−0.005***	−0.004***	−0.002**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
PerceivedPrice	0.009***	−0.000	0.008*	0.007*	0.010***	0.003	0.001
	(0.003)	(0.004)	(0.004)	(0.004)	(0.003)	(0.002)	(0.002)
Hotel FE	YES	YES	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES	YES	YES
District-specific trends	YES	YES	YES	YES	YES	YES	YES
Observations	73,159	64,066	60,113	64,879	74,268	84,406	85,647
R-squared	0.193	0.228	0.200	0.228	0.221	0.174	0.174

Note: Clustered robust standard errors in parentheses.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

home sharing's entry, which is contrary to the prediction of the alternative explanation.

4.3.2 | Matched sample using propensity score matching (PSM)

To alleviate concerns regarding the comparability between the affected hotels and unaffected hotels, we conduct a robustness check by combining a PSM method with DID. PSM helps address the selection concern by accounting for observables that predict a subject receiving the treatment, that is, to ensure the affected and unaffected groups are comparable before the treatment (Abadie & Imbens, 2006). Specifically, we run a logistic regression at the hotel level using the mean values of observables to predict the probability of a hotel experiencing home sharing's entry. These observables include hotel price segment (i.e., whether it is higher-priced), previous guest satisfaction, and sales performance, as well as the number of competitor hotel cohorts in its vicinity. We then construct an unaffected hotel group using those that mirror the affected hotels on all observable attributes but have not yet experienced home sharing's entry. As a result, we have matched 1171 hotels from the treated group and 1171 hotels from the control groups. We present the balance check of covariates of the PSM sample in Section W3 of the Web Appendix (Table W4). The covariates are quite balanced from the matching. There is no significant difference in covariates between the two groups. Using the PSM sample, we replicate the main analysis and report the results in Section W3 of the Web Appendix (Table W5). The results are highly consistent with those reported in the main specifications.

4.3.3 | Look-ahead (LA) PSM

Some may still worry about the comparability between the affected hotels and unaffected hotels. To further account for observable and unobserved characteristics, we adopt an LA-PSM method to identify the proper control group (Bapna et al., 2018, Kumar et al., 2018). As illustrated in Figure W4, the main idea of LA-PSM is to use only treated units in the sample and compare those treated earlier with those treated later. The benefit of an LA-PSM model is two-fold. On the one hand, the closest propensity score ensures that the treated and the control units are similar in observable characteristics; on the other hand, choosing hotels from those that have not experienced home sharing's entry in period 1 but will experience in period 2 ensures that the treated and the control units are similar in unobserved characteristics. We

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TABLE 9 Impact of hotel management responses on negative reviews by service areas

D.V.: log*NegativeReviews* on Review Topics	(1) T1 Check-in/out	(2) T2 Facility and amenity	(3) T3 Cleanliness	(4) T4 Sightseeing experience	(5) T5 Room conditions	(6) T6 Location	(7) T7 Guest service
T1 MR Ratio $(t-1)$	−0.224***						
	(0.019)						
T2 MR Ratio $(t-1)$		−0.081***					
		(0.018)					
T3 MR Ratio $(t-1)$			−0.128***				
			(0.019)				
T4 MR Ratio $(t-1)$				−0.178***			
				(0.025)			
T5 MR Ratio $(t-1)$					−0.104***		
					(0.020)		
T6 MR Ratio $(t-1)$						−0.159***	
						(0.021)	
T7 MR Ratio $(t-1)$							−0.116***
							(0.018)
Control Variables							
logHotels	−0.058	−0.086**	−0.085**	−0.019	−0.054	−0.133***	−0.069
	(0.044)	(0.040)	(0.043)	(0.059)	(0.047)	(0.044)	(0.042)
Rating	0.461***	0.429***	0.407***	0.293***	0.355***	0.496***	0.438***
	(0.063)	(0.065)	(0.074)	(0.090)	(0.077)	(0.070)	(0.075)
logReviews	0.082***	0.038***	0.064***	0.067***	0.045***	0.072***	0.064***
	(0.010)	(0.010)	(0.011)	(0.012)	(0.012)	(0.011)	(0.011)
PerceivedPrice	0.107***	0.074***	0.091***	0.040***	0.057***	0.093***	0.077***
	(0.012)	(0.012)	(0.013)	(0.015)	(0.013)	(0.012)	(0.012)
Hotel FE	YES	YES	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES	YES	YES
District-specific trends	YES	YES	YES	YES	YES	YES	YES
Observations	22,169	20,778	20,157	16,894	18,443	20,481	19,951
R-squared	0.739	0.715	0.700	0.597	0.645	0.676	0.718

Note: Clustered robust standard errors in parentheses.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

re-estimate Equation (1) using the new LA-PSM sample. The results are presented in Section W4 of the Web Appendix (Tables W6 and W7) and are consistent with those in the DID and PSM-adjusted DID models.

4.3.4 | Instrumental variable approach

Some may worry about the endogeneity issues in estimating the home-sharing impacts. We adopt an instrumental variable approach using a measure of changes in home-sharing reviews as the instrument to mitigate the endogeneity concern. In particular, the instrument is the logged total number of reviews on home-sharing properties within a hotel's vicinity in a given month (log *Home Share Reviews*). This measure meets both the relevance condition and the exclusion condition to be a valid instrument (Roberts & Whited, 2012). First, log *Home Share Reviews* is likely correlated with the growth of home sharing. If the supply of home-sharing properties increases in a hotel's vicinity, it is likely there will be more home-sharing reviews (a positive correlation between log *Supply* and log *Home Share Reviews*). Second, log *Home Share Reviews* are randomly contributed by home-sharing travelers and should not be related to hotel performance. It is unlikely that home-sharing reviews directly affect hotel management responses and performance if not through the home-sharing supply. Taken together, log *Home Share Reviews* is an appropriate variable to instrument the estimated impact. The results are presented in Section W5 of the Web Appendix (Table W8) and remain consistent with

TABLE 10 Impact of hotel management response absence on negative reviews by service areas

D.V.: log*NegativeReviews* on Topics	(1) T1 Check-in/out	(2) T2 Facility and amenity	(3) T3 Cleanliness	(4) T4 Sightseeing experience	(5) T5 Room conditions	(6) T6 Location	(7) T7 Guest service
T1 No MR $(t-1)$	0.042						
	(0.028)						
T2 No MR $(t-1)$		−0.003					
		(0.021)					
T3 No MR $(t-1)$			0.015				
			(0.024)				
T4 No MR $(t-1)$				0.030			
				(0.037)			
T5 No MR $(t-1)$					−0.009		
					(0.028)		
T6 No MR $(t$-1)						0.037	
						(0.031)	
T7 No MR $(t-1)$							0.027
							(0.028)
Control Variables							
logHotels	−0.051	−0.074**	−0.078**	−0.078	−0.073*	−0.128***	−0.085**
	(0.042)	(0.037)	(0.039)	(0.051)	(0.042)	(0.043)	(0.040)
Rating	0.422***	0.348***	0.421***	0.256***	0.283***	0.334***	0.300***
	(0.055)	(0.049)	(0.060)	(0.066)	(0.057)	(0.057)	(0.058)
logReviews	0.083***	0.040***	0.054***	0.060***	0.049***	0.068***	0.062***
	(0.010)	(0.009)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)
PerceivedPrice	0.105***	0.081***	0.087***	0.034***	0.066***	0.098***	0.079***
	(0.012)	(0.011)	(0.012)	(0.013)	(0.012)	(0.012)	(0.011)
Hotel FE	YES	YES	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES	YES	YES
District-specific trends	YES	YES	YES	YES	YES	YES	YES
Observations	22,169	20,778	20,157	16,894	18,443	20,481	19,951
R-squared	0.722	0.711	0.693	0.578	0.636	0.658	0.703

Note: Clustered robust standard errors in parentheses.
$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

the main specification, lending support to the robustness of the findings.

4.3.5 | Alternative treatment variable

In main analyses, we observe home sharing's entry by investigating the supply of home-sharing properties in a hotel's vicinity of a 500-m radius. Some may wonder if the results hold robust using a different measure of home sharing's entry. Instead of using the continuous measure of home-sharing properties, $logSupply_{it}$, we use a binary variable $Entry_{it}$, to indicate whether home sharing has entered a hotel's vicinity (our treatment variable). The results are reported in Section W6 of the Web Appendix (Table W9). All our estimates remain qualitatively consistent as the main analysis.

4.3.6 | Alternative home sharing entry measure

Some may wonder if the results hold robust using different radii to define a hotel's vicinity. We compute the vicinity of a hotel using the 1000 and the 3000-m radius and repeat our analyses. The results, as exhibited in Section W7 of the Web Appendix (Tables W10 and W11), are consistent with our main specification using the 500-m radius.

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4.3.7 | Alternative sample with price similarity

Some may expect that home sharing is for budget-conscious guests, and such a unique guest base of home sharing would not affect hotels in general. We have shown that home sharing attracts guests at comparable price segments with hotels (see Figure 1 for a price comparison between home sharing and hotels). It is unlikely that home sharing is just for budget-conscious guests. Nevertheless, to further address this concern, we identify home-sharing properties falling in the price range of +20% and −20% of a given hotel, *logSupplyPrice*20%. We use this alternative home sharing property sample to re-run Equation (1). Section W8 of the Web Appendix reports the estimation results (Table W12). Specifically, the findings suggest that the impact of home sharing holds robust as it clearly competes with hotels even at a similar price range (i.e., −20% and +20% of hotel price).

4.3.8 | Alternative differentiation of hotel groups

Some may wonder if the divergence in hotel responses and the performance outcome after home sharing's entry will remain consistent using the alternative differentiation of hotel groups. Instead of using the hotel price to create the binary measure of *HighPrice*, we use the hotel star level (Nie et al., 2022) to classify hotels. In particular, we generate a binary variable of *HighStar*, with the value of 1 indicating a hotel is a scale level above the median value of hotel star levels and 0 otherwise. The results are reported in Section W9 of the Web Appendix (Table W13), which are consistent with our main results using the price segment binary variable.

5 | IMPLICATIONS AND CONCLUDING REMARKS

Sharing economy services are disrupting incumbent firms by cutting into their customer base and sales. With rapid and accelerating market changes, incumbents need to adapt and react to the competition proactively. In this paper, we explore whether and how incumbent firms can align their strategies with these challenges. While the majority of the literature focuses on price strategy when facing entrants, we propose a non-price, quality improvement strategy of management responses toward performance improvement. Our results show that hotels that actively respond to guest reviews significantly improve guest satisfaction and sales. This is because management responses are effective in identifying service areas in which hotels fall behind home sharing so that hotels that actively respond to guest reviews can bridge the gaps for improved performance against the competition.

On substantive contributions, this paper makes one of the first attempts to investigate management responses as a non-price strategy in responding to sharing economy competition and the associated mechanism behind the performance improvement. It systematically adds to the emerging literature on incumbents and home sharing, the impact of management responses in competition, and deep learning on non-structural data (e.g., guest reviews). More specifically, it generates new insights on several issues that are important but less explored in the literature, including how incumbents become adaptable when facing disruptions using non-price strategies, the duo effects of incumbents in not only addressing competition from external disruptors but also differentiating the incumbents from peer cohort incumbents, and how deep learning can be deployed at the firm level to assist addressing competition strategies.

On practical contributions, this paper taps into the external environment of incumbent firms and advocates a non-price strategy as the pace of competition accelerates. Despite the industry landscape being constantly driven by technology disruption, we advocate that the incumbents should take a guest-centric approach to quality improvement rather than lowering prices simply to disruptors. One of the strategies advocated in this study is for managers to respond to (and learn from) the ubiquitously available guest reviews for service areas to improve—especially areas in which quality gaps between incumbents and disruptors exist. We show how incumbents can utilize management responses to exploit performance improvement opportunities in guest reviews and sustain competitive advantage facing increased competition from home sharing. Through responding to reviews and automating deep learning, different types of incumbents can extrapolate and scale the efficiency for actionable strategies specific to their improvement needs while remaining competitive in the market.

Our hybrid method integrating causal inference and deep learning is theoretically and practically relevant. We first see this study as one of the recent attempts to address unstructured data, rapid speed of generation, and the large volume of information (i.e., the "big data" problems) in business research. Despite the impact of guest reviews on business performance has been well established in the literature (Chevalier & Mayzlin, 2006), less work has focused on insights from review contents (beyond review ratings) that speak to the specific performance improvement opportunities against disruption. Through systematically transforming reviews into quantifiable outcomes, this paper adds to the emerging literature that lies at the intersection of operations management and technology. Second, the advantage of our deep learning algorithms is that they let us sift these theme-specific content features (topics and sentiments) across a wide range of guest reviews without hand-coding features with human interventions or domain knowledge. If automated and deployed by incumbent businesses, this approach can not only extract cues in guest reviews for performance improvement opportunities but also monitor the guest sentiments for competitors. Leveraging the power of deep learning, the managers can proactively identify areas to bridge the quality gaps toward elevated guest satisfaction and sales performance, thus battling back against the disruption from home sharing.

The limitations of our study point to several future research directions. First, dynamic pricing is an important feature of hospitability services. Because we do not have time-variant price data, we focus primarily on the non-price strategy (i.e., management responses) in responding to disruptors without combining both price and non-price strategies. It will be intriguing to compare both price and non-price strategies of hotels across price segments. Second, topic and sentiment analyses evolve rapidly, with new approaches such as attention modeling and aspect extraction available to make algorithmic processing more effective. We use a method with high accuracy and reliability following the literature (Hollenbeck, 2018; Liu et al., 2019; Xu et al., 2021) without incorporating these new, more advanced approaches. Future studies can explore various methods that can improve the efficiency and accuracy of the algorithmic processing, especially on un-structural textual review data. Notwithstanding these limitations, we believe that our study serves as a stepping stone for further research along the line of improving service operation management given the ever-changing technologic disruption.

ACKNOWLEDGMENTS

The authors thank the department editor Vijay Mookerjee, the senior editor, and three anonymous referees for their thoughtful suggestions throughout the review process. For helpful comments, the authors are grateful to the reviewers of the NET Institute Grant 2019 and the Marketing Science Institute (MSI) Research Grant Competition 2019, as well as the reviewers, discussants, and participants at the 2019 Conference on Artificial Intelligence, Machine Learning, and Business Analytics, NET Institute Conference 2019, INFORMS Annual Meeting 2019, Statistical Challenges in Electronic Commerce Research 2019, and Marketing Dynamics Conference 2019. The study is partially supported by the NET Institute Summer Grant and the MSI Research Grant. Jianwei Liu acknowledges the support of the National Natural Science Foundation of China (72002058, 72131005) and the Hong Kong Polytechnic University (P0034898). Karen Xie, Wei Chen, Yong Liu, and Yunlong Sun did not receive any support from the National Natural Science Foundation of China (72002058, 72131005) and the Hong Kong Polytechnic University (P0034898), and were not involved in any of the funding applications.

ENDNOTES

[1] Source: http://pages.ctrip.com/public/ctripab/abctrip.htm.

[2] Source: http://www.xiaozhu.com/aboutus.

[3] We also measure the entry of home sharing using the radius of 1000 and 3000 m of a hotel to check the robustness of our results. The estimations are consistent with the main specification using the 500-m radius. We report the analyses in Web Appendix A5.

[4] We provide more justification for this proxy in Section 4.1 when we discuss the model specification.

[5] We measure the perceived hotel price based on the sentiments detected in guest reviews related to price. First, we randomly select 10,000 hotel reviews and hire evaluators to mark (a) whether these reviews are price-related and, if yes, (b) what is the reviewer's sentiment (negative, neutral, and positive) toward the hotel price. We then train a classifier using the evaluators' labels to classify all the hotel reviews and calculate the monthly ratio of positive price-related reviews to all the price-related reviews for each hotel (i.e., *PerceivedPrice*). Our classification accuracy is 91.52%. The specific classification process is similar to what is described in Section 4.2.

[6] We also use a PSM method with DID to check the robustness of our findings. The result is consistent as reported in Web Appendix W2.

[7] We also check the robustness of the results using a dummy variable, $Entry_{it}$, to indicate whether home sharing has entered the 500-m vicinity of a hotel. The estimation reveals qualitatively consistent findings with our main specification and is reported in Appendix W4.

[8] We also use alternative hotel vicinities such as the 1000 and 3000-m radius to check the robustness of our results. The findings are consistent with our main specification and are reported in Web Appendix W7.

[9] To see this, both $Treated_i \times logSupply_{it}$ and $logSupply_{it}$ have values of zero in the control group. In the treatment group, they are both zeros before home sharing's entry because $logSupply_{it}$ is zero in this period. After home sharing's entry, both are equal to $logSupply_{it}$ because $Treated_i$ is always one for the treatment group, and $Treated_i \times logSupply_{it}$ becomes $logSupply_{it}$.

[10] *MatchedMRRatioDiff* measures the difference in ratios of management responses with contents that are similar to guest reviews. Rather than the cookie-cutter management reviews, we regard management responses with contents that are more similar to (or more relevant to) guest reviews as those of higher quality. It is calculated via deep learning techniques. We first conduct text mining on the content of hotel management responses and guest reviews using a unified LDA model (the detailed training processing is introduced in Section 4.2 and Section W2 in Web Appendix). We then count the number of topic overlaps (or matches) between guest reviews and management responses at the hotel level. We further calculate a ratio of managerial responses that have matched topics (with guest reviews) to all management responses of for a given hotel.

[11] $MRRatioDiff_i$ and $MatchedMRRatioDiff_i$ will be absorbed by the hotel fixed effects because they are time-invariant differences before and after home sharing's entry for each hotel, as described in Section 3.

[12] The magnitude of change in ratings may be too small to be significant. The distribution of ratings is relatively centralized, with a mean of 4.214 and a median of 4.38. The actual changes are at a small level.

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How to cite this article: Liu, J., Xie, K., Chen, W., Liu, Y., & Sun, Y. (2023). How incumbents beat disruption? Evidence from hotel responses to home sharing. *Production and Operations Management*, *32,* 2758–2774. https://doi.org/10.1111/poms.14005

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